

Academic Performance Task, Expectations and Rubric

The Task

Students begin by displaying the dominant land cover in each ecozone and interpret its value as wildlife habitat. The initial inquiry will focus on the distribution of bird, mammal, plant, and reptile species across Canada. Students will be expected to examine potential reasons for the various species distributions they encountered. Students will then sum the reptile & amphibian, tree, rare plants, endemic plants, bird and mammal species numbers for each ecozone to get a total number of species in each ecozone. This will allow students to determine if there is a correlation between species at risk and certain ecozones, and what might be contributing to the species being at risk. Finally students will use existing hot linked text files within ArcView and the Internet, access the <http://www.speciesatrisk.gc.ca/Species/English/SearchRequest.cfm> web site, to select a species at risk. A brief report is to be prepared which provides information as to why the species is currently at risk and what is / can be done to improve the habitat for that species. The report should conclude by suggesting ways to prepare for the future. Students are instructed to include supporting visuals, such as species distribution maps, graphs, charts, diagrams, or pictures, in the article.

Expectations

This task gives students the opportunity to demonstrate achievement of the following selected expectations from three strands: Geographic Foundations: Space and Systems; Human – Environment Interactions; and Methods of Geographic Inquiry.

Students will:

1. demonstrate an understanding of the characteristics of natural systems (e.g., climate, landforms, soils, natural vegetation, wildlife);
2. demonstrate an understanding of how natural and human systems interact within ecozones;
3. analyse the ways in which natural systems interact with human systems, then make predictions about the outcomes of these interactions;
4. explain how the effects of urban growth (e.g., development on former farmlands, destruction of wildlife habitats, draining of marshes) alter the natural environment;
5. select and use appropriate methods for displaying geographic data;
6. create and use effectively photographs, charts, graphs, models, and diagrams;
7. use geographic data to support conclusions and opinions.

Prior Knowledge and Skills

To complete this task, students are expected to be able to demonstrate the following:

- knowledge of the definition of an ecozone;
- knowledge of some of Canada's ecozones and their physical characteristics (e.g. landforms, climate, vegetation, and soils);
- of what value are wild animals to Canadians?;
- knowledge of wilderness areas of Canada and human and animal needs for wilderness;
- understand the differences between extinct species, extirpated species, endangered species, threatened species, at-risk species, and vulnerable species;
- understanding of research methods using a variety of source materials;
- ability to read, analyse, and interpret graphs and maps.

Task Rubric – Species at Risk						
Expectations	Criteria	Level 1 (50 – 59)	Level 2 (60 –69)	Level 3 (70 – 79)	Level 4 (80 -100)	
<u>Knowledge / understanding</u>						
1, 2	- Demonstrate an understanding of the characteristics of natural systems (e.g., climate, landforms, soils, natural vegetation, wildlife); - Demonstrate an understanding of how natural and human systems interact within ecozones	Demonstrates little understanding of how natural and human systems interact within ecozones	Demonstrates some understanding of how natural and human systems interact within ecozones	Demonstrates considerable understanding of how natural and human systems interact within ecozones	Demonstrates a high degree of understanding of how natural and human systems interact within ecozones	
<u>Thinking / Inquiry</u>						
3	Analyse the ways in which natural systems interact with human systems, then make predictions about the outcomes of these interactions	- Uses a limited variety of relevant supporting facts and details - Explains the significance to a limited degree	- Uses some variety of relevant supporting facts and details - Explains the significance to some degree	- Uses a considerable variety of relevant supporting facts and details - Explains the significance to a considerable degree	- Uses a wide variety of relevant supporting facts and details - Explains the significance to a high degree	
<u>Communication</u>						
5,6,7	- Use visuals to help illustrate the importance of the issue - Write a clear and convincing report	- Uses visuals to illustrate the importance to a limited degree - Writes a report that is not very clear or convincing	- Uses visuals to illustrate the importance to some degree - Writes a report that is somewhat clear and convincing	- Uses visuals to illustrate the importance to a considerable degree - Writes a report that is generally clear and convincing	- Uses visuals to illustrate the importance to a high degree - Writes a report that is very clear and convincing	
<u>Application</u>						
4	Explain how the effects of urban growth alter the natural environment	Shows limited understanding of the implications	Shows some understanding of the implications	Shows considerable understanding of the implications	Shows thorough understanding of the implications	